

Enterprise Information Integration

The New Way to Leverage E-information

Second Edition

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Aberdeen Group, Inc.

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Preface

Enterprise information integration (EII) is now a market separate from enterprise application integration (EAI) and extract-transform-load (ETL) solutions.

The main reason that companies now recognize a distinct need to integrate information across the enterprise is that the overall pool of proprietary enterprise information — as distinct from information associated with specific packaged applications, Web databases, or data mining sources — is now a key e-business differentiator. The ability to leverage information from a wide variety of storage types, environments, applications, databases, and files via enterprise portals allows enterprises to deliver major value-add simultaneously to employees, partners, and customers.

A second reason to look closely at EII is that — as Aberdeen research shows — firms are now at a critical stage in linking back-end applications and data to their front-end Web sites. Three years of fragmented efforts to link key applications one by one, followed by a proliferation of enterprise portals that crisscross various back-end data sources, have left companies with unnecessarily complex and costly e-business architectures. The key to cutting administrative and new application development costs is to implement a common software infrastructure architecture — and EII infrastructure lies at the core of such an architecture.

Third, the requirements of e-business are a superset of the requirements of traditional information processing. A narrow definition of data integration as being concerned only with integrating information *within* a company must change to encompass EII's integration of information *between* companies as well.

This report provides the reader with a clear understanding of the forces influencing the market, as well as Aberdeen's insights into the real opportunities and benefits of EII and its future directions. The second edition follows up the information published in May 2002 to include major new suppliers and new trends. The report includes the following:

- *Technology segmentation* — An update to the Aberdeen segmentation model that indicates various technology approaches to EII and the suppliers supporting those technologies
- *Market size and growth projections* — Aberdeen's most recent data on EII market size and growth

- *Issues and challenges* — An outline of additional industry trends that are affecting EII initiatives
- *Observations and conclusions*
- *Supplier profiles* — A comprehensive list of the suppliers in this market

The intended audience for this report is IT (information technology), corporate, and line-of-business management responsible for planning and managing initiatives for information integration, as well as communicating and interacting effectively with potential distributors and suppliers. This publication was designed to help these users understand the potential impact of information integration regarding the following:

- An enterprise's total application portfolio
- Enterprise information strategies
- E-business planning and execution
- Present technologies, including Web services, which will underpin corporate architectures over the next two to three years

This publication profiles the key suppliers and products in the market. Aberdeen believes that this report will aid organizations in identifying the technologies and attributes of EII products that can best fit their business requirements. It will also help organizations effectively plan EII acquisition strategies.

Chapter One:

Executive Summary

Aberdeen research shows that the motivator of many e-enterprises is not only cost savings but also competitive advantage. A slight Web site advantage can cause customers across a global market to establish relationships; a slight but persistent disadvantage can, even today, cause 10% to 20% of the installed base to walk away.

To create and maintain competitive advantage, enterprises have now moved beyond pretty graphics, informational displays, and rudimentary order taking. They have also connected their back-end order-processing (B-to-C) and supply chain (B-to-B) applications to the Web site — and, in some cases, with other companies' Web sites or with e-marketplaces. However, most of these efforts maintain separate, uncoordinated links to one or more Web sites.

To integrate information at the user-interface level, enterprises are now implementing enterprise portals. A portal allows the enterprise to display data applications and fully leverage all major business-critical company databases. However, by itself, an enterprise portal does little beyond show data in different formats side by side.

To fully leverage the connections between information from different applications and databases, an enterprise needs an EII infrastructure to aggregate data and coordinate transactions across back-end data sources. At its most powerful, EII lets all back-end information be seen as if it came from one comprehensive, global database.

The resultant Web site-based informational applications, or *e-information*, do more than show the latest company news. As with portals, they can serve as free advertising, attracting the information researcher who is hungry to find a one-stop shop. They can complement customer relationship management systems by providing the information desired at each point in the customer's buying decision process. Within an organization, they can tie the company together and ensure faster, better decision-making. In some cases, such as stock market reports and analysis, they can serve as money generators in their own right.

Along with these competitive advantages come built-in cost savings — not only the cost savings from doing everything over the Internet, outsourced or consolidated, but also the IT cost savings from simplifying the administration of the database/application connections and, thereby, saving key administrative costs. Moreover, users report that developers writing to a common high-level database interface save

significant time in both development and upgrading data-intensive applications that cross data sources.

Of equal importance is a side benefit: e-information can unify the enterprise's archipelagoes of data. Whereas data warehouses failed to persuade data fiefdoms to merge, the Web site manager now has the political power to enforce common data standards on company divisions that seek the selling power of its Web site — but only if that manager plans ahead via an e-information strategy.

A further note: Wringing Web site-displayed informational applications, or e-information, from existing databases is not easy. Hence, competitive advantage is more durable and differentiation is easier.

An early example of EII's potential comes from the field of employee self-service applications. EII plus an enterprise portal delivers an immediate impact to the bottom line by eliminating the need for "information intermediaries" that stand between the individual employee and the information that he or she needs.

The benefits of employee productivity EII do not end with the bottom line. For example, employee-productivity solutions based on EII provide additional information to employees, such as company policies and standards or information about the company's products, and they can deliver critical cross-database information in a just-in-time fashion. This information, in turn, can give users the ability to make higher quality, more rapid decisions because they can base their decision making on higher quality data that is more up-to-date. Visualization and personalization of the information can improve the employee's comprehension of the data. Improved accuracy of the information can reduce the risk that company employees — or the company itself — will violate government regulations.

However, one of the main potential advantages of EII is not yet on users' radar screens — the ability to provide a common administrative locus for all major enterprise databases. Aberdeen research shows that database administration costs now dominate the TCO (total cost of ownership) of applications below the 500-user level, and they continue to increase in importance for all sizes of applications. Users correctly identify the ability to manage multiple databases with one common interface as a major source of cost savings, but they often look to application servers as a method to achieve cross-database administration. However, full-featured EII, with its greater knowledge of database content, is much better suited to provide database administration not only across a few applications at the Web site but also across many or all databases within the enterprise and even outside of it.

In Chapter Three, Aberdeen describes how EII suppliers are meeting the challenge of delivering a "virtual federated database." Aberdeen

finds that providing a scalable, robust, low-administrative-cost solution is a tough nut to crack, but today's EII solutions already provide strong value-add by allowing the identification of new cross-database, cross-application, and even extra-enterprise sources of information and data interrelationships. Today's suppliers are, on the average, larger and better able to provide supporting services than a year ago, so users no longer need fear the perils of being pioneers.

Chapter Two:

Summary of Findings

- Currently, EII is a small market — less than \$200 million in size — that lacks visibility to IT buyers. Over the last year, however, EII visibility has grown dramatically; this growth is especially due to IBM's announcement of its Information Integrator products.
- Buyers are beginning to see EII as a strategic infrastructure decision that can improve the agility of IT and enhance IT's ability to quickly adapt technology to the rapidly changing needs of the business — especially the needs of e-business.
- The business benefit of EII increases exponentially with successive integration of each related application and database. The connections between each additional information source and all the others have typically not been examined in most enterprises, and these connections — especially when combined with business process software — allow employee empowerment, greater insights into competitive advantage, more rapid response to customer inquiries and orders, and more effective integration with the databases of supply chain partners.
- EII solutions can enable and support today's three high-priority e-business initiatives — customer relationship management, e-business integration across the supply chain, and Web-based employee productivity applications. Moreover, they offer a way to speed up business-process-integration efforts by allowing simultaneous access to multiple databases attached to multiple applications.
- Partial deployment of intranet, extranet, and Internet applications without EII increases the difficulties of implementing EII, but also increases the costs of failing to implement EII by increasing the costs of administering unnecessarily complex architectures.
- The EII market is still new, and most EII suppliers are small ones, such as MetaMatrix, Venetica, XAware, Enosys, and Nimble Technologies. Over the past year, large firms have begun adding the technology — BEA is contracting with Enosys to provide EII as part of BEA's Liquid Data for WebLogic initiative, and IBM's Xperanto EII project has resulted in the new IBM DB2 Information Integrator products.

- EII-type functionality is a cornerstone of today's more successful enterprise portals and of the Web services that enterprises are beginning to build. Initial EII implementations are project specific and seek to simplify developer interfaces in which portals, Web services, and similar software makes cross-database queries.
- Many EII suppliers have experience in data integration and data warehouse ETL. These suppliers' experience in optimizing data-transmission performance across databases means that today's EII solutions are by no means bleeding edge; in fact, they often offer the robustness and scalability of mature technologies. The advent of larger suppliers — such as BEA, Business Objects, and IBM — ensures that IT buyers will run fewer risks of supplier failure or inadequate service.
- The potential of the EII market will mean that major hardware and database suppliers will continue to enter the market within the next two years, either by acquisition or by partnerships with existing EII suppliers. The entry of these suppliers will continue to increase EII's visibility, allowing them to parlay advanced technology-based solutions and industry relationships into large-scale enterprise opportunities. If successful, these suppliers will fuel a \$250 million-plus market over the next two years. Increasingly, EII solutions will be incorporated in portal, EAI, business intelligence, and business-process-integration solutions, making EII a key technology in a \$7.5 billion market.
- Over the next two years, EII will begin to form part of a larger "information aggregation" market aimed at delivering tools for strategic, proactive information management. EII will be a key enabler for identifying, managing, and leveraging competitive-advantage proprietary information and information relationships.
- EII can play an effective role in exposing enterprise data to outside suppliers in supply chain and e-business integration solutions.
- The strength of industry partnerships — professional services and systems integrator firms, enterprise portal and application suppliers, and system infrastructure suppliers — and of the consulting that EII suppliers provide on how to best utilize EII is as critical to EII implementers' success as the technology that the suppliers offer. IT evaluators should be looking especially to those EII suppliers that continually enhance their capabilities across all three of these dimensions.

- One key EII supplier technology differentiator is the degree to which a supplier uses XML (Extensible Markup Language) technology for inter-database integration. The XML approach makes it easier to deliver the resulting aggregated data to a wide variety of inter- and extra-enterprise audiences. Another is support for SOAP and Web services.
- Finding the appropriate skills to execute an EII project is not always easy, as EII implementation requires in-depth knowledge of the structure of existing enterprise databases. This will provide a business opportunity for professional service suppliers and for application service providers (ASPs) that offer EII integration services.
- Evaluators should favor EII solutions that allow users to define the databases to be integrated and provide developer-friendly user interfaces for doing so; support a wide variety of databases; and provide customized support for “de facto standard” application interfaces (e.g., enterprise portals) and delivery mechanisms (e.g., XML).
- At present, no EII solution offers administration across data sources. However, this capability is a key value-add to many users. EII’s ability to drive down database administration costs will reduce per-application TCO significantly.
- In many organizations, the same team that monitors the server farm should make the EII buying decision. This requirement stems from the fact that understanding the enterprise’s e-business architecture is key to effective implementation of EII and new applications that leverage EII.

To fully reap the benefits of EII, evaluators should also apply the following assessment criteria:

- *Extensibility and reusability* — The integration process should be extensible and reusable for the project at hand, as well as for future projects. The tools should be designed in such a manner that users can extend the tool from a per-project basis to an ongoing enterprisewide usage.
- *Flexibility* — The solution should be able to fit into an existing architecture and handle new technologies without massive rewrites and without involvement from the supplier.
- *Efficiency* — The solution should be capable of operating across a wide range of performance levels (e.g., very fast, fast, moder-

ate, and overnight) and integration models (e.g., asynchronous messaging and synchronous transaction).

- *Interoperability and breadth* — The solution should understand the relationships between various database technologies and provide interfaces to interdependent middleware technologies such as application servers.
- *Cost-effectiveness* — In the long term, the solution should be able to contribute to an organization's bottom line by reducing the cost and risk of developing and maintaining custom, one-off, EII solutions. The upside value of a common repository of information for all major enterprise databases is immeasurable.
- *Ease of maintenance* — In EII projects that involve the integration of many data sources, maintaining independently coded modules for each integration process is extremely resource intensive and, hence, costly. Using a single, flexible, and consistent development framework for each process will greatly reduce the amount of personnel resources needed to maintain and develop the system.
- *Deployment ease and efficiency* — Deploying the runtime environment should be easy, and the EII solution should allow deployment to a wide variety of target systems, ensuring optimal use of the enterprise network. Upgrade deployment should also be simple.
- *Ease of administration* — EII modules will be distributed widely across an enterprise — just as enterprise-packaged application modules are today — and conceivably across enterprises where a supply chain relationship exists. These integration modules will need to be managed and administered with no less diligence than is applied to current systems, networks, databases, and applications.
- *Industry acceptability* — A large number of partners should incorporate the EII technology and deliver solutions based on it.
- *Technological innovation* — The solution should provide and fully exploit the latest integration technology.

Chapter Three:

Market Definition and Dynamics

What Is EII, and How Does It Work?

EII is software that provides a “database veneer” that makes disparate data sources appear as one database. It allows users to apply data management transactions to the pooled data, at a corporate or enterprise level, in order to support applications that manage, present, or analyze the data in new ways.

Currently, EII has three key technical characteristics:

- It is a *software infrastructure*. It is not an application; it supports the creation of applications, such as enterprise portals, that display and analyze combined data. As software infrastructure, EII forms a key part of any overall e-business architecture.
- It *links the second and third tiers* of a typical enterprise architecture in a “many-to-one-to-many” approach. In other words, EII gives a wide variety of second-tier Web applications running on the second tier of an e-business architecture access to multiple back-end and legacy databases and file systems running on the third tier.
- It provides an *enterprise-scale, integrated* approach. Unlike today’s point-to-point application-to-database links, EII provides a common infrastructure on which all links are built, incrementally if the customer wishes. This common infrastructure can deliver more rapid development, more cost-effective centralized administration of data from multiple data sources, and more flexible presentation and analysis of more data sources.

All EII products usually offer certain basic components:

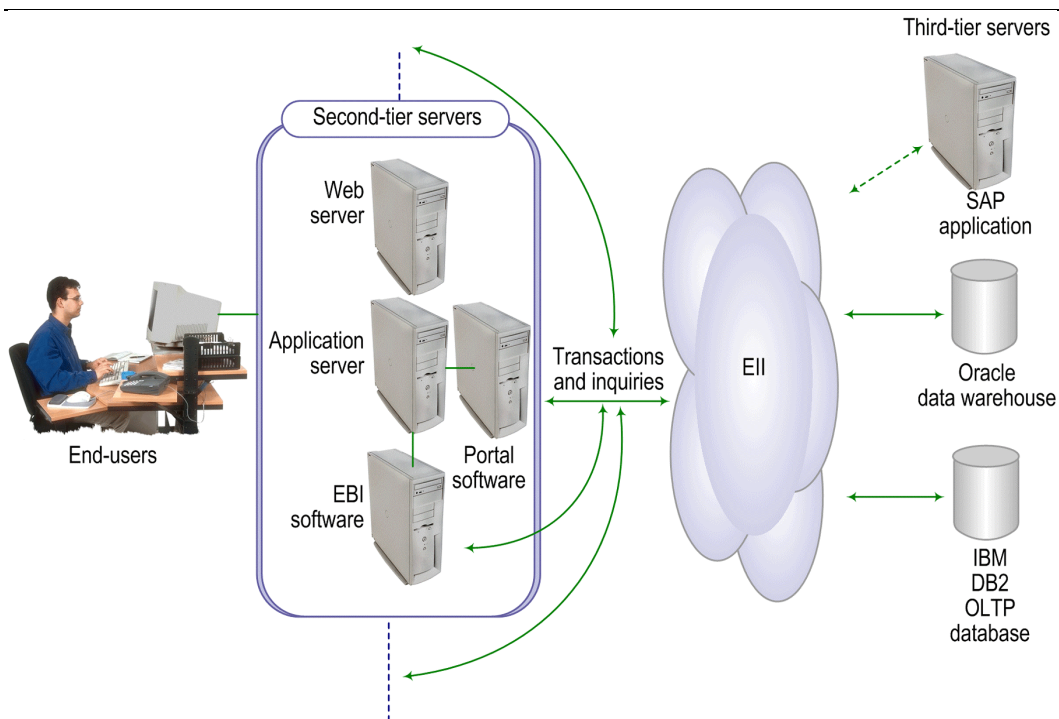
- *A directory* — This “database about databases” contains information about the data stored in the databases and often contains the same information that is stored in each database’s data dictionary, as well as information about the data that is shared across databases. A directory may also include information about “normalized” versions of the data — i.e., data converted into a common format that is particularly suitable for passing on to users, Web sites, or enterprise portals.
- *Conversion capabilities* — An EII solution must often convert incoming data from nonrelational to relational format or from relational to object format. When updates to back-end databases

are involved in a transaction, an EII solution must also perform the “inverse” task of converting a common format back into the target database’s format. Note that current EII solutions that support updates do so under strict conditions and/or the user’s control.

- *A database veneer* — To the user, the EII solution should look like a database and, if possible, to the developer and administrator as well. Therefore, the EII solution should at least support standard SQL and querying transactions.
- *Front- and back-end communications* — The EII solution should have a way of communicating with front-end programs and with users via programmatic and communications invocations. It also should be able to send transactions to and receive results from back-end databases.

Figure 1 shows how EII fits into a typical architecture.

Figure 1: EII in a Typical Architecture



Source: Aberdeen Group, July 2003

EII Is New Technology

The IT buyer should carefully distinguish an EII solution from an ETL engine for data warehousing, an EAI solution, or a content management system. An ETL engine (sometimes called a “data integration” solution) allows multiple streams of data to be converted to a common data warehouse data format and be loaded into the warehouse “in bulk” or in short “bursts.” However, it does not deliver that data to external targets, such as an enterprise portal; it does not make the source data appear as one database; and it does not typically emphasize transactional access to the data.

An EAI solution passes data among multiple packaged applications. It does not deliver that data to other external targets, such as a Web site or enterprise portal, and it often operates at the level-of-business processes rather than at the data level.

A content management system allows users to place a wide variety of rich-media data into a common database and to operate on that data as if it were a single database. The content management system requires that data be uploaded to a common database, whereas the EII solution leaves that data in place. The typical content management system focuses on a variety of data, such as text, rich media, and business rules, whereas most EII solutions currently focus on relational operations that act upon relational and nonrelational data.

EII in the User Site

EII solutions serve three masters: the developer, the administrator, and the user.

In order to serve the *developer*, the EII solution provides a bridge between the components and business logic residing in the second tier of the e-business architecture and back-end data sources — that is, it allows the developer to treat multiple, disparate databases, data management systems, and files as one gigantic enterprisewide database.

To aid the *administrator*, the EII solution often (but not always) provides a “metadirectory” of data and object types (e.g., an XML data dictionary) across multiple data sources, plus specific tools that allow the administrator to monitor, maintain, and implement security for the virtual federated database.

To aid the *user*, the EII supports the kind of operations on the virtual federated database commonly found in SQL (Structured Query Language), usually querying (decision support) and sometimes add/remove/update (online transaction processing [OLTP]). These features allow support for e-commerce and B-to-B OLTP, enable Web analytics queries across databases, and abet enterprise portal

display and a combination of information from various data sources for employees, customers, and partners.

Situations in Which Today's Companies Often Feel a Need for EII

Aberdeen clients find that the data connectivity projects for which EII is a new tool are no small tasks. Typically, the project scope is greater than anticipated, and the data connectivity problems involve heterogeneous environments. IT administrators usually must perform these integration projects in a complex environment and must rely on a rapidly changing technology infrastructure. Other frequently encountered issues include the following:

- A hodgepodge of legacy systems
- Multiple legacy databases from multiple suppliers
- Disparate operating platform environments — MVS, Unix, Windows NT and XP, wide area network (WAN), local area network (LAN), and Internet
- Heterogeneous and geographically dispersed database management and file systems
- Many communications protocols, including SNA, TCP/IP, HTTP, and XML/SOAP

Finally, there is no guarantee that the data-level integration will have integrated the applications' business processes as a seamless whole when the project is done. Today's data connectivity solutions deal with transactions on a per-application, per-database basis, and they do not make it easy to handle business processes that involve different databases or different applications.

By contrast, EII:

- Supports transactions across databases and applications, making the implementation of business processes much easier
- Makes the reuse of integration techniques across heterogeneous environments much easier
- Gives users the ability to implement integration solutions without needing detailed knowledge of the application infrastructure's specific underlying technologies.

Thus, for all of these issues and situations, EII provides added simplicity leading to lower costs.

Business Benefits of EII

Aberdeen research shows that EII solutions can deliver not only cost savings but also a new kind of competitive advantage. As it becomes more difficult for e-enterprises to achieve differentiation via packaged or custom applications, the new frontier of competitive advantage becomes an IT organization's ability to leverage proprietary information that its competitors cannot access. Much of this information is scattered across multiple databases, spreadsheets, and rich-media Web sites, making it difficult, if not impossible, to detect the connection between a customer's existing product preferences and new products arriving via the supply chain and R&D. EII's ability to amass metadata and carry out transactions across data sources allows the EII user to detect and display these key "relationships across data frontiers."

EII's cost savings are nothing to sneeze at either. Many of today's tasks in BI (business intelligence), CRM (customer relationship management), portals, and consolidation involve integrating business processes and information. In these cases, EII simplifies the developer's job by giving one interface and one set of metadata for all data sources, speeding the development of applications such as enterprise portals. EII simplifies the administrator's job by exposing key data that must be kept in synchronization across data sources.

The benefits that EII delivers to the enterprise are often immediate. The ability to cross database boundaries, if used by applications, rapidly allows wider user access to key existing data for data mining, enterprise portals, or Web searches from inside or outside of the enterprise. Establishing relationships between data sets across frontiers and data source types quickly enhances the data warehouse, empowering deeper data analysis and providing a more comprehensive view of the enterprise's information assets. The development of today's business-critical, data-intensive applications is speeded when programmers need only code for access to one data source rather than to several disparate ones. Key administrative costs show significant improvement the moment that administrators can manage data more easily across data-source frontiers.

In real-world implementations, users are increasingly seeing specific instances of the following benefits:

- EII gives a broader picture of data to the user; e.g., by combining data in Oracle and IBM DB2 databases.
- EII encourages IT administrators to identify data copies and related data across databases, files, spreadsheets, and graphics and video. For the user, this means new insights into a company's proprietary information — for example, an alert that a new

product in which the customer has an interest is about to arrive and can be pre-ordered. For the administrator, this means a better understanding of the enterprise's overall data relationships, so that, for example, the administrator can ensure that changes in one set of data changes other related data appropriately.

- EII simplifies the developer's job. Instead of writing custom code for each Java access to back-end data, the developer can write once for all accesses, whether they involve access to a single database or multiple ones. By combining EII with Web services, the developer can simplify the front end of the application (by making it a Web service) and the back end (by simplifying data access). By "abstracting" the interface to back-end databases, EII makes it easier to upgrade an application — there is "one application programming interface (API) to choke" when data access must be changed.
- Although no EII products currently support this process, it should be possible to allow administrators to create scripts that perform database administration across databases and file systems — for example, scheduling offline backup of Oracle and Microsoft databases in sequence, to load balance a backup.

Potentially, EII can be used as part of an overall IT strategy to identify, manage, and leverage the enterprise's proprietary information for competitive advantage (strategic information management). For a full description of this idea, see Chapter Four.

Applying EII to Typical IT Tasks

When deftly used, a powerful EII tool can make a significant contribution to many of an IT organization's most critical tasks (Table 1).

Table 1: Ways to Use EII in Strategic Initiatives

Aim	Way to Use EII
Integrate business processes	Write code in the applications to access an EII solution rather than the databases so that the EII can determine if the transactions are part of a business process and update all data sources at the same time
Implement Web services interfaces to existing information	Write a Web services front end to an EII solution rather than to each data source
Speed time to react to current customer or supply chain data	Provide an EII interface to the data for query purposes, bypassing the database server

Aim	Way to Use EII
Speed time to develop new applications that leverage proprietary information, such as portals	Use an EII solution to front end for both the data warehouse and other proprietary information, ensuring one set of code for all information access Use an EII solution's XML support to avoid coding object-to-relational translations or access to unstructured data
Audit all information in the enterprise for government requirements	Create an EII metadata repository by importing (or generating) metadata from the data warehouse, ETL tools, and file systems and rich-media sites, and then write a SQL or Xquery query to the EII interface
Cut IT costs	Cut administrative costs by creating an administrative "veneer" over an EII tool that issues administrative commands to multiple data sources Cut development costs with an EII tool that allows developers to "write once" to supply data access to many applications
Merge companies	Combine the metadata of the two companies via an EII metadata repository
Increase the amount of information available to users	Create EII across multiple databases that expands the range of information available to an enterprise portal via the same interface

Source: Aberdeen Group, July 2003

The following real-world examples show that EII is making a contribution:

- Merrill Lynch uses MetaMatrix to speed development and to allow analysts to access stock, product, and pricing data in IBM, Oracle, and Sybase databases.
- Aventis uses IBM to mine its Oracle databases and external nonrelational sources (genomics, proteomics) for drug research, and they are seeing significant speedups in R&D.
- Crystal Decisions uses IBM as an ISV to improve performance in accessing large amounts of heterogeneous, distributed data in thousands of data sources via Crystal Reports; it claims that response time has improved by up to 98%.

Developments in the Database Market Drive the Need for EII

The IT buyer should consider another reason for assessing EII tools: the state of the database, EBI (enterprise business integration), and portal markets.

Periodically, database suppliers attempt to “own” the market by persuading organizations to migrate existing data to their database. Organizations are perfectly willing to do so — after all, the cost of administering multiple databases is a major and increasing component of a summed IT total cost of ownership. However, moving all key enterprise data onto Oracle, for example, has almost always proven impossible to achieve. Legacy databases will continue to live.

As a result, organizations need a second-best alternative: to administer, as far as possible, multiple databases as if they were from one supplier. This alternative not only dramatically simplifies the job of the database administrator but also eases the process of incorporating a wide variety of data (spreadsheets, data in the form of text, multimedia data) into the enterprise’s visible information portfolio. Thus, a cross-database-supplier solution sharply reduces administrative time required (and, hence, costs), and enlarges the scope of the data administered.

At the same time, the need to aggregate data across databases and deliver it to a wide variety of audiences is increasingly urgent. The two key drivers of the market’s demand for cross-database solutions are EBI and enterprise portals. As XML-based solutions proliferate, they create a demand for more varied back-end data to feed to outside suppliers. Enterprise portals, by their nature, seek to display a wide variety of data on a single screen, and ODBC- and JDBC-based solutions simply require too much work of the programmer to be implemented easily.

To meet these needs and overcome the difficulties associated with existing solutions, EII acts as a many-to-many interface between the user and multiple databases. Table 2 shows how this approach allows the EII supplier to overcome the difficulties that have faced past approaches.

Table 2: EII versus Traditional Approaches

Single “Central” Database”	Programmatic Interface (JDBC)	EII
Cannot keep up with upgrades to other databases	Can keep up only with upgrades common to all databases	Can keep up with upgrades common to all databases, allows programming to covert new data types to a standard format, and allows users to bypass the EII solution for key new single-database features
No programming interface for cross-database-supplier transactions	Low-level programming interface for cross-database-supplier transactions	High-level components and programming interface (in some cases, looking like a single database) for cross-database-supplier transactions
No provision for handling data defined in multiple databases	No provision for handling data defined in multiple databases	Directory that “tags” data that exists in multiple databases
No load balancing for transactions across multiple databases	No provision for load balancing for transactions across multiple databases	In many cases, built-in load balancing for transactions across multiple databases

Source: Aberdeen Group, July 2003

Criteria for Buying an EII Solution

EII solutions differ markedly in their ability to support developers, users, and administrators. Aberdeen recommends that IT buyers use different criteria according to which of these is the “main constituency” for an e-business architecture. The following are key criteria for developers:

- *Scalability* of the development tools and of the applications built over the EII solution. Development tools should allow team and remote development (e.g., via a “developer network”); data-access mechanisms should provide optimization of queries and transactions.

- *Quality assurance* to ensure robust e-business applications (e.g., by providing the ability to persist data from the start of a cross-database transaction to the end, transparent to the developer).
- *Programmer productivity*, by componentizing key EII operations at the business level, allowing the developer to swiftly write EII-using applications, such as enterprise portals.
- *Flexibility* to support a wide variety of platforms and architectures, by supporting open standards such as J2EE (Java 2 Enterprise Edition), EJBs (Enterprise JavaBeans), and XML and by offering the ability to add new platforms and architectures easily and rapidly.

The following are key criteria for administrators:

- *Comprehensive cross-database administration* features that allow a database administrator to monitor and maintain multiple databases via EII as if they were one database, as well as coordinate with the database administration tools of back-end and legacy databases
- *Ability to monitor and tune transactions* within the EII system in order to detect bottlenecks, preferably via a remote console
- *Connections to application-management and systems management products*, such as CA Unicenter TNG

The following are key criteria for users:

- *Comprehensive support* for a wide variety of back-end databases and other data sources
- *Easy “plugability”* into front end e-commerce, B-to-B, and, especially, enterprise portal implementations, as well as popular Web and application servers
- *Powerful cross-database transactional features*, such as sophisticated querying

IT buyers should also consider the following overall criteria:

- *The supplier’s ability to serve as a “one-stop storefront”* for system design and integration. The IT buyer should be especially attentive and avoid “finger pointing” (the case in which multiple suppliers blame each other for a failure) because EII solutions typically involve complex connections to a wide variety of supplier software.

- *An EII solution's ability to be a good foundation for a Web services architecture.* Over the next two years, Web services implementations will become much more frequent, and EII must provide the standards support to integrate well with other upcoming Web services architecture components, such as a UDDI-based directory, as well as have the ability to access a wide variety of intra- and extra-enterprise data sources that these Web services promise.

Critical Technology Factors

Aberdeen research has determined that the following five factors are key to understanding and differentiating EII solutions:

- Object and multimedia data support (including XML support)
- Non-database information access
- Specific support for front-end users and applications such as browser-based analytics, EBI, or enterprise portals
- Developer support
- Administrator support

Object, Multimedia, and Non-database Support

Data integration solutions will have a variety of constituents — internal programs, Web site users, workgroups within the enterprise, partners/suppliers, internal workgroups or intranets via a LAN (local area network) or remotely, data miners, and IT administrators. These constituents vary widely in how they are accustomed to having data presented (Table 3). A good EII solution is, therefore, *many-to-many*: It draws data from a wide variety of data types, including object and multimedia ones, and delivers the data in a standardized format supported by each constituent.

XML is the coming lingua franca (or common language) that allows this type of flexible response to a wide variety of constituents. Wrapping a transaction result in XML allows it to be delivered not only within the organization, where IT administrators can enforce XML support, but also to outside users via the Web, as well as partners and suppliers that are increasingly looking to XML for electronic data interchange and EBI.

The EII solution should also support access to text files, spreadsheets, graphics, video, and objects, as well as relational data. Increasingly, business-critical data includes at least one of these non-traditional data types.

Table 3: EII Constituents and Their Data Needs

Constituent	Data Typically Accessed	Data Format Typically Required
Internal programs	Relational	Relational
Web site users	Text, graphics, relational	HTML
Partners/suppliers	Relational, text	XML
Internal workgroups or intranets	Spreadsheet, text	HTML, graphical
Data miners	Relational	Graphical
Administrators	All types	No requirements

Source: Aberdeen Group, July 2003

Most EII suppliers are moderately advanced in supplying these features. Most support access to a wide variety of data, although the emphasis is usually on relational data. Most also have some variant of XML support either available or planned. Few provide specific support for a broad range of constituents.

Specific Front-End Support

Specific types of new solutions that many enterprises are implementing call for specific support from an EII solution. The following are major new application types that EII should support:

- Enterprise portals
- EBI
- Browser-based analytics

Enterprise Portals

Today's enterprise portal, via a rapid, straightforward implementation, allows the coordinated presentation and use of a much broader range of information to B-to-C and B-to-B customers, as well as to internal intranets and data miners. However, these connections work best if integrated with the whole infrastructure of a Web's architecture. That includes the following:

- *Middleware* — e.g., Web servers, application servers, application messaging, and EAI
- *Databases*, as both second-tier caches and as “common” information stores
- *Development tools* and frameworks for rapid development and customization
- *Hooks* to integrate B-to-C and B-to-B supplier order processing
- *XML formatting* for Net market display
- *Decision-support querying*

The EII solution stands between the enterprise portal and “common information store” databases, acting as a collection point for the data in order to forward it to the enterprise portal for display. The EII solution can also help with second-tier data caching, portal development, XML support, and querying across databases. In order to do so effectively, the EII solution can offer a caching database, its own development tools that are integrated with those of the enterprise’s portal-infrastructure solution, XML support, and its own querying facilities.

EBI

The EII solution can also serve to bypass the enterprise portal in presenting single-database and cross-database corporate information to partners and suppliers. By avoiding the user interface characteristic of an enterprise portal, the EII solution can simplify both many-database-to-one supplier and many-database-to-many-supplier interactions. However, this also means that the EII solution must supply the XML-standard support, tools for building connections on the supplier side, and security features characteristic of today’s EBI solutions.

Browser-Based Analytics

The promise of browser-based analytics is to extend the information analysis benefits of today’s data warehouses to a wider range of Web-based users and to simplify analysis so that it is not the province of a few highly skilled data miners. An EII solution can increase the scope and improve the ability to analyze real-time information of an analytics solution by allowing querying across a wide variety of data, not simply the data in a corporate data warehouse. To support browser-based analytics, EII solutions should supply some of the characteristics of a browser-based analytics solution in a cross-database situation — governors to avoid the “query from hell” that crowds out all other users from accessing a database, cost optimizers

to maximize cross-database query performance, and tools to convert back-end spreadsheet and relational data to graphical (pie chart, statistical) representations.

Most suppliers now provide specific features for browser-based analytics and portals, with fewer targeting EBI.

Developer Support

For both the developer and the administrator, the ideal EII solution provides a “single database image,” in the same way that a good clustering solution pretends that multiple systems are a “single-system image.” This, in turn, means that the EII solution provides most of the developer features of a high-end database — stored procedures, standard SQL support, a wide variety of APIs (including support for ODBC, JDBC, and EJBs), and the ability to use the EII solution’s directory to drive program design (e.g., by mapping directory “metadata” into common design tools such as Oracle’s and Rational’s design solutions). Note that the EII solution cannot offer one of the advantages of the single database — the ability to run a program inside the database’s process for high performance.

Most EII solutions are fairly far advanced in providing the developer support that users need. Many solutions support stored procedures, and they all support standard SQL. Some provide development tools bundled with the application. Relatively few offer a full array of high-end database APIs, and few provide mapping between the metadata directory and design tools.

Administrator Support

Again, the EII solution should provide an image of a single database to the administrator. However, this goal is much more difficult to achieve for administrator support than for developer support. It effectively means that the EII solution must offer cross-database backup/recovery, cross-database security, and interfaces to each database’s own administrative tools for functions such as upgrade, reorganization, and replication. The EII solution should also offer SNMP (Simple Network Management Protocol) support to allow the EII administrative tool to interact with common systems management tools, such as CA Unicenter TNG.

Aberdeen characterizes the current administrator support tools offered by EII suppliers as workable but basic. A few suppliers have tools for transaction monitoring, user management, and security. What is especially missing from all suppliers today is support for the integration of these administrative tools with enterprise system and network management facilities such as Unicenter TNG, Tivoli, or OpenView. Current EII solutions are likely to be widely deployed in

large-scale and highly distributed computing environments, so it is obvious that enterprise administrators will need to be able to manage EII nodes and modules just as they do distributed EBA (enterprise business application) system components.

Notably, few, if any, EII suppliers provide support for cross-database administration scripting.

Chapter Four:

EII As Part of Strategic Information Management

The proliferation of files, rich media, and traditional relational data, as enterprise computing broadens in the number of users supported and IT value-add, has become an inexorable long-term trend. Ways to access this information are hindered by the differences in how this information is stored, to the point where each Oracle database, group of Excel spreadsheets, or Web site can require a separate program to elicit data-based insights. Thus, the “frontiers” between information sources delay competitive-advantage data analysis, add cost to the process of creating queries and transactions across data boundaries, and mire today’s efforts to leverage proprietary information into long-term revenue enhancement in a hopeless game of “catch-up” with enterprises’ growing cache of information.

Over the years, specialized tools and technologies have arrived that tackle the problem of information frontiers. These technologies have now reached a point of maturity in which the IT executive can use them strategically, in what Aberdeen calls *strategic information management*. The cross-frontier technologies that support strategic information management are what Aberdeen calls *information aggregation technologies*.

As an information aggregation tool, EII has an important role to play in strategic information management, in concert with other such tools. This chapter catalogs the other technologies that constitute information aggregation, shows how EII and other information aggregation tools can be used (singly or in combination) to abet strategic information management, and examines the state of the art that IT strategists can use today in attempting strategic information management.

A Brief Introduction to Strategic Information Management

Today’s IT strategist, seeking to increase competitive advantage, can focus primarily on the application resources or the data resources of the enterprise. Until recently, it made most sense to focus on applications. *Strategic solution management* has sought to identify, manage, leverage, and add to the application software that delivers value-add to the enterprise, customer, or supply chain.

Over the last two years, IT organizations’ focus has shifted strongly toward cost cutting. Cost squeezes at both ISVs (independent service

vendors) and major IT shops have nearly halted the development of new, competitive-advantage applications or major Web efforts. Meanwhile, CIOs and ISV CEOs have resorted to cost-saving measures, such as offshore development outsourcing. However, strategies based solely on cost savings typically reach a point of no return — the ultimate result of long-term cost cutting is a “hollowed-out” corporation.

Today, the obvious place to search for a cost-effective, long-term advantage is, therefore, in proprietary information that competitors cannot readily duplicate. For example, Amazon.com may seek to combine information about customers’ preferences with information about upcoming book releases, notifying them when a new book of interest is about to be published. However, this information is usually locked in widely differing and often aged data sources. In another example, much Wall Street investment research depends on data from sources that range from Excel spreadsheets to reports, up-to-the-minute news articles, Web pages, and historical stock databases. Simply writing new code or buying a packaged business intelligence application does not do the trick. The IT strategist also needs to focus on making the information more widely available; therefore, strategic information management is needed.

Strategic information management has a short- and a long-term component. The short-term job of strategic information management is to deliver available information consistently, no matter the variations in user demand — “on-demand” information. Tools that support the short-term job include storage management and SAN (storage area network)/NAS (network-attached storage) management (especially virtualization), database and cross-database administration, system and systems file administration, and Web site administration.

The long-term job of strategic information management is to identify, manage, support the leveraging of, and add to the information resources of the enterprise. Tools that support the long-term job(s) include metadata repositories and cross-database administration. Tools that support the “information aggregation” discussed here are just beginning to surface, and tools that allow automated searches of sources of new proprietary or extra-enterprise information that can be used for competitive advantage are not yet visible in the market.

What Is Information Aggregation, and Why Does It Matter?

Information aggregation is Aberdeen’s term for a collection of tools and technologies that have arrived over the last seven years and that allow users to combine data across data-source frontiers. Information aggregation includes the following:

- Tools to move or copy data from multiple “source” data sources to a “target” aggregated-data repository, which can be used to carry out such operations as queries across differing data types
- Tools to access data from multiple data sources and then combine it temporarily for specific purposes, such as cross-database queries

Aberdeen arbitrarily calls this “information aggregation” rather than “information integration” for two reasons. First, compared with the term “integration,” “aggregation” better captures the idea that data can be combined both permanently and temporarily. Second, “information integration” may be confused with “enterprise information integration” — a new sub-technology of information aggregation — as well as various recent efforts by suppliers to coin the terms data integration, enterprise application integration, and information integration.

Information aggregation currently includes the following types of tools:

- Data migration and bulk download tools
- Replication and ETL tools
- Data warehouses and data marts
- Enterprise operational data stores
- Mid-tier operational data stores
- EII tools
- Federated databases

Any information integration technology delivers immediate benefits to the enterprise. The ability to cross database boundaries used by applications gives wider user access to key existing data for data mining, enterprise portals, or Web searches from inside or outside the enterprise. Establishing relationships between data sets across frontiers and types of data sources empowers deeper data analysis, leverages proprietary information more effectively for competitive advantage, and provides a more comprehensive view of the enterprise’s information assets. Development is speeded when programmers need only to code for access to one data source rather than several disparate ones. Administrators often cut costs by managing more easily across data-source frontiers.

When IT administrators wish to manage information strategically, information aggregation can play a key role in leveraging existing

information. Information aggregation's ability to establish relationships between data sets in widely different data sources helps the IT strategist identify key data and key "data connections" by placing that "information about information" in one place. Moreover, typical side effects of information aggregation tool use are the gathering of cross-data-source metadata and the ability to administer aggregated data and metadata. Thus, the information management strategist can

Table 4: Strategic Uses of Information Aggregation

Key Task	Value-Add of Information Aggregation
Identify	ETL tools and data warehouses can create a cross-database metadata repository. EII tools can add file and rich-media data-source metadata.
Manage	Data warehouses and operational data stores, fed by data-movement tools (data migration, data cache, ETL), allow full administrative operations across at least one copy of aggregated relational data sources and, sometimes, file and rich-media data sources. EII and federated databases potentially allow full administrative operations across all copies of relational, file, and rich-media data sources.
Leverage	Data warehouse, federated database, and EII APIs allow broader, more quickly developed application access to relational data sources. EII extends application access to file and rich-media data sources. Enterprise portals extend application access to a broader range of users. Operational data stores, federated databases, and EII can allow update as well as query access.
Find new use	No effect (still a manual process)

Source: Aberdeen Group, July 2003

also identify and manage information across frontiers more effectively. Table 4 suggests specific ways in which specific information aggregation tools can be useful in each key strategic information management task.

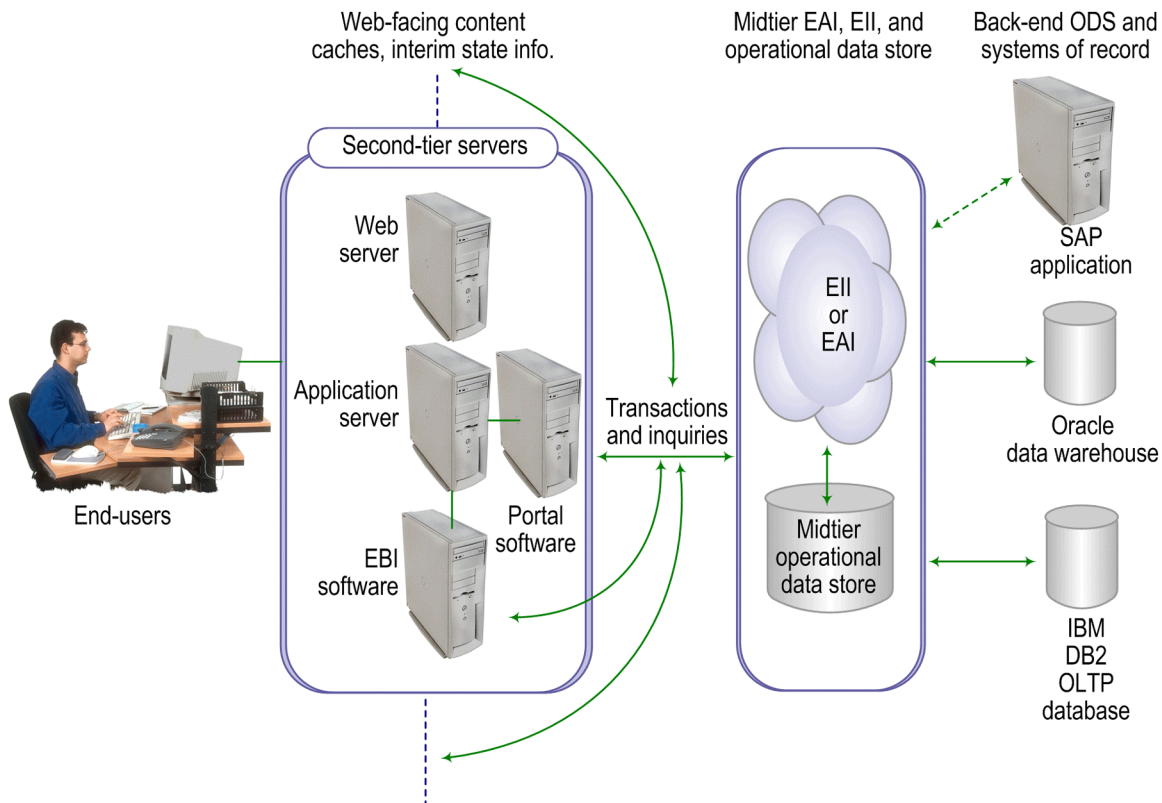
The culmination of these benefits occurs when an IT shop decides to implement full-fledged strategic information management. By applying each information aggregation tool appropriately and strategically, the IT strategist can potentially attain an "almost enter-

prisewide” metadata repository, an almost enterprisewide data-source administration tool, and an almost enterprisewide information-leveraging tool to constantly search the enterprise’s ever-growing information store for new ways to combine the information for competitive advantage.

When to Apply What Information Aggregation Technology

Figure 2 shows how information aggregation technologies can be combined in an overall architecture.

Figure 2: Example of Information Aggregation Architecture



Source: Raining Data Inc. and Aberdeen Group, March 2003

The State of the Art in Information Aggregation

Data Migration and Mass Download Tools

Data migration and mass download tools were the first information aggregation tools, and they linger in today's ETL products that can also be used for bulk data migration and download. For example, Evolutionary Technologies Inc. (ETI) continues to provide data migration capabilities in its "data integration" product.

Moreover, the enterprise-database-supplier consolidation in the late 1990s spawned several tools for migrating from one database to another — e.g., from Sybase and Informix to IBM relational databases

or from these databases to Oracle. In addition, several major enterprise database suppliers offer tools to migrate files and/or rich-media “objects” into their relational databases — e.g., Oracle and IBM — or to move or copy large amounts of other data — e.g., IBM to move IMS data into a DB2 database.

Most of these tools focus on migration from one relational database to another. The general-purpose data migration and ETL tools are the most flexible, allowing migration across suppliers and, in some cases, across a wide array of data types. Therefore, they are often preferred for long-term strategic information management because they can be reapplied in many different cases in which permanent data movement is desired or in which database consolidation will likely be ongoing.

Table 5 shows one way to categorize information aggregation technologies.

Table 5: Taxonomy of Information Aggregation Tools

Tool	Characteristics	Typical Uses Today
Data migration and mass download	Typically <i>move</i> large amounts of data at once, either permanently or intermittently, almost always “offline” (when the target data store is not operating) Typically one source, one target	End use of a database by moving it to another database (e.g., move to another database supplier) Allow database operations (especially queries) to be applied to a data source (e.g., PC files) by moving/copying it inside the database Backup and disaster recovery

Tool	Characteristics	Typical Uses Today
Replication and ETL tools	<i>Copy</i> (one way) one piece of data at a time, either continuously as data changes or in spurts, usually offline Transform data to different format to improve its quality or to convert to a common format Typically multiple sources, one target	Refresh a data warehouse or data mart from mission-critical OLTP (online transaction processing) relational databases, typically daily Reflect local-office data to a central site for backup and querying (“mass-deployment”) Synchronize handheld/laptop and central databases Copy data between embedded databases for packaged applications from different suppliers to integrate business processes (EAI)
Data warehouses and data marts	<i>Copy</i> (one way) large amounts of data at a time, usually daily, almost always offline Typically multiple sources, one target Usually includes ETL tool Allow querying of aggregated data, but not update Often extremely large (terabytes)	Support querying, data mining, and business intelligence Provide metadata repository of key business information
Enterprise operational data stores	<i>Copy</i> (one way), typically changes in data (immediately) — often <i>from</i> the store/cache to multiple targets, always online Typically multiple sources, one target Often use replication or ETL tool Allow querying and update Often large, but not so large as data warehouse	Combine key customer and supply chain data to detect and query relationships Combine data from key business applications to integrate business processes Improve speed-to-react to business conditions (refreshed more frequently than data warehouse) Improve visibility of current (less than a week old) data

Tool	Characteristics	Typical Uses Today
Mid-tier operational data stores	Copy (one way) changes in data (immediately) — often <i>from</i> the store/cache to multiple targets, always online Typically multiple sources, one target Often use replication or ETL tool Allow querying and update Smaller than enterprise operational data store	Combine key customer and supply chain data to detect and query relationships Combine data from key business applications to integrate business processes Improve speed-to-react to business conditions (refreshed more frequently than data warehouse) Improve visibility of current (less than a week old) data Improve performance and scalability of key applications and “composite” applications
EII	Leaves the data sources as is Often uses real-time ETL tool Allows querying and (less frequently) update Must transform data for each query	Queries and updates across a wide variety of data sources (including file and rich media) without creating copies to synchronize Creates a more global meta-data repository than any other IA tool Allows cross-data-source development Potentially, allows cross-data-source administration
Federated database	Leaves the databases as is Allows querying and update Must transform data for each query	Combines differing relational databases from the same supplier or two suppliers

Source: Aberdeen Group, July 2003

Replication and ETL Tools

If an IT organization chooses to copy data from a data source to data “target,” then changes in that data must likewise be copied from source to target. If an IT organization needs up-to-date sources and targets at all times, the data copies must be kept in synchronization — every transaction must see the same value for the data at all times, no matter what copy is changed first. In OLTP situations, that means using two-phase commit technology (a technical method of achieving two-way synchronization between two information bases). The drawback of this technology (as with all two-way synchronization technologies) is that it is low performance — it can take a long time to carry out a single multiple-copy update, much less a high-volume stream of them. Thus, in synchronized OLTP systems, only a few data copies are realistically possible.

In data warehousing environments, the situation is simpler. The data warehouse is not intended as an OLTP environment; it is in no hurry to record changes to data items. Therefore, all a data warehouse needs is one-way synchronization, or data *replication*. The OLTP “feeder” databases record a change in the data and, at some later point, it is replicated to the copies in the data warehouse database.

Of course, even in a data warehousing context, replication is not without cost. Updates that occur while the data warehouse is running means write locks that can slow querying. Selective updates of some data items, although related data items are not updated until the end of the day, may result in inconsistent data. However, these problems are typically easy to handle — slower querying by delaying updates until times of low querying (such as the end of the day) — enabling mass replication to update data all at once and inconsistent data to be avoided by careful data warehouse design. Deft use of replication, therefore, can result in far more timely data on which data miners can operate.

Replication tools abound in the market, and the major database suppliers offer many of them. These tools may allow replication from multiple suppliers’ relational databases to a particular supplier’s database, but not vice versa. However, because it is easy to find another replication tool to perform “reverse replication,” one-way replication is not a major drawback.

ETL tools combine both data migration and replication tool technology. As with data migration technology, they allow the translation of data formats from widely different source formats into widely different target formats via a process of extracting, transforming (with cleansing), and loading. As with replication tools, they can copy data in real time and in small amounts. An ETL tool can be combined with replication technology to ensure that when data is changed, that change is sent on to the target database.

Initially, ETL tools were targeted primarily at the data warehousing market, but the advent of EAI has meant that ETL tools are also packaged as part of EAI solutions, often as “adapters” that translate between the data formats of major packaged applications such as those provided by SAP and PeopleSoft. Recently, ETL tools have appeared in the offerings of EII suppliers.

The wide array of ETL tools in the market is often from suppliers that specialize in data warehousing add-ons, or EAI, or both — suppliers such as Compuware, Data Junction, ETI, Sybase, TIBCO, and webMethods. IT strategists should note that ETL tools are exceptionally flexible and, therefore, applicable as the base technology for a wide variety of information aggregation needs.

Data Warehouses and Data Marts

Data warehouses aggregate data primarily, and often exclusively, for decision support. A data warehouse’s data is derived from outside OLTP and “mixed” databases. The databases’ data is fed to the decision-support information base via either mass downloads involving careful cleansing of the data (typically via an ETL tool) or by one-way synchronization, in which changes in specific data items are replicated in the data warehouse information base. Specialized querying or analytics tools allow specialized “data miners” to access the data and derive insights about patterns of sales or similar competitive-advantage information. Data marts are smaller data warehouses — often for specific functions within an enterprise — that feed a large central data warehouse.

Experience has shown that the data warehouse has major limitations that prevent it from serving as the ultimate information aggregation of all data in the enterprise. First, it can grow only to a certain size before performance becomes too poor; hundreds of terabytes may sound like a large number, but the typical major organization stores far more data than that. Second, in order to service the “query from hell,” the data warehouse must focus on queries to the exclusion of all other operations, including the updates that accompany key operations, such as customer order entry. Third, holders of line-of-business data or other data are often reluctant to register it in the data warehouse’s metadata repository because it brings no immediate business benefit. Fourth, refreshing a large data warehouse daily is a massive operation that prevents warehouse usage for hours, so keeping the warehouse more than one day or one week behind is, practically speaking, impossible.

Nevertheless, the data warehouse is often the best base on which to begin implementing strategic information management. It is not only the largest aggregation of data in the enterprise; it is also the largest aggregation of metadata.

All major enterprise database suppliers — IBM, Microsoft, Oracle, and Sybase — offer extensive data warehousing support, as does NCR-Teradata.

Enterprise Operational Data Store

The idea of the operational data store is to combine the key data from multiple data sources or “systems of record” in one database that — unlike a data warehouse — allows the updating of the data. An enterprise operational data store focuses on collecting business-wide, mission-critical data. This approach gives the enterprise operational data store the ability to aggregate the data of a data warehouse and provide near real-time “data freshness,” giving users the ability to react to changes in business conditions — such as shifts in buying patterns — instantaneously. However, the price of adding updates is a sharp decrease in the amount of data that can be stored in the operational data store and the amount of sophisticated data mining that can be applied to it. In other words, in an enterprisewide operational data store, users can react more quickly to insights, but gain fewer of them.

All enterprise databases can be used as enterprise operational data stores when combined appropriately with real-time ETL or replication tools. The trick to successfully implementing an operational data store is not so much picking the right database as picking the right data to go into the database. However, the process of picking the right data is precisely what makes either an enterprise or a mid-tier operational data store such a valuable tool for strategic information management. All data is not equal; the strategic information manager needs to distinguish proprietary data that can be leveraged, and that type of data is frequently found in an operational data store.

The idea of an enterprise operational data store, pioneered by suppliers such as Unisys, is now generally accepted across the industry, and most major enterprise database suppliers now offer support for implementation of enterprise operational data stores (often referred to simply as “operational data stores”). Note that many enterprise operational data stores combine only data from relational databases.

Mid-tier Operational Data Stores

A mid-tier operational data store is a recent idea with distinct merits for strategic information management. A high-performance database serves as a persistent, long-term cache, not only for one but also for several databases from one or several suppliers. This “cache” database, unlike most caches, provides all of the functionality of an operational data store. This functionality allows users not only to speed the performance of each data source but also to combine key data from each data source, reflecting changes to this key data back to all data sources. In other words, a mid-tier operational data store can

provide all of the benefits of an enterprise operational data store (although often on a smaller scale), as well as improved performance and scalability for mission-critical databases and applications.

As with EII and EAI tools, the enterprise and mid-tier operational data stores can store metadata about relationships across databases and, therefore, provide users, administrators, and developers with the benefits of an EII solution on a much smaller scale. In addition, EII and operational data stores can be combined quite effectively. Recent examples of mid-tier operational data stores include Raining Data's TigerLogic XML Data Management Server (XDMS) and Ipedo Information Hub.

EII Tools

The potential for EII as an enabler of strategic information management cannot be overestimated. EII can serve as the focal point for gathering an unprecedented amount of metadata about the enterprise's proprietary information. Potential EII tools can administer across as wide a swath of the enterprise's data sources as desired. EII allows leveraging data relationships across data sources without the effort of moving or copying the data.

The Usefulness of Combining Mid-tier ODS and EII

Users should note that EII tools typically do not include any persistence technology, so information that is extracted, transformed, and integrated for each request can cause performance overhead compared with an ODS — although the ODS does not permit access to as large a data store as an EII tool. Thus, a combination of EII and an operational data store is often a better combination of data-store size and performance than either would be separately.

This is true not only for decision-support querying but also for tasks such as order entry that require data updates. A mid-tier ODS's ability to index data from other data sources allows users to not only query these sources — an ability that all EII tools share — but also insert, remove, and update them (actions that relatively few EII tools permit).

Moreover, a combined EII/mid-tier ODS lets the information strategist “have it your way” — (1) to merge data into an ODS, where swift access to real-time data for rapid decision making is paramount; (2) apply a cross-data-source query rather than moving data, when an ODS “tops out” or operational data must be combined with data warehouse historical data; or (3) rapidly react to new floods of data by moving some into the ODS and access the rest via EII.

Federated Database

The federated database may well be an idea whose time has passed; however, it is still worth noting as an alternative approach to information aggregation that may yet pan out. The idea of a federated database is to provide the same data management software to different data sources. Thus, in Informix's federated approach, the same version of Informix OnLine accesses object and relational data previously served by different Informix databases, allowing the same queries, administration, etc., to be used on both data sets. Other variants would allow the same operations across multiple suppliers' data sources or across widely different data sources.

The major difficulty so far with this one-size-fits-all approach is that it has proved very hard to make one operation fit all data sources — e.g., to make query optimization appropriate to relational data effective in optimizing queries of video-heavy large Web objects. That is compounded by the fact that the amount and types of data in the typical enterprise are continually proliferating, so the federated database implementer is chasing a moving target.

Periodically, an enterprise database supplier will propose to get around this problem by simply providing different “sub-database” software within the same database for each data type, then persuading the organization to migrate all other data in the enterprise to its database. Three years ago, it looked as if Oracle might actually achieve this goal — but that was a market mirage. Data migration of all legacy databases and upgrading of all legacy applications, depending on each legacy database, is simply too hard for most if not all enterprises.

The result is that, for now, the vision of one enterprise-standard database or of a federated database that can encompass all data sources is pretty much dead. Organizations can still apply a federated database to combine two existing data sources from the same supplier, if that supplier supports it, but that is only a minor step toward information aggregation.

Action Items

Clearly, today's information aggregation tools have enormous potential to support strategic information management by helping to identify, manage, and leverage existing proprietary competitive-advantage data. Just as clearly, no one has combined these tools in an overall tool set to do this.

The action item for IT buyers, therefore, is to assess existing information aggregation tools within their enterprises, supplement the tools where necessary, and create a long-range plan to use them to identify, manage, and leverage the enterprise's data sources. A key

to success in this effort is the implementation of EII not merely on a per-project basis, but also as an ongoing effort to amass metadata about all enterprise data sources, including nonrelational ones. The political payoff for “data fiefdoms” that provide this metadata is greater exposure to customers via the Web and greater cost cutting as database administrators operate across data sources.

The action item for suppliers is to actually implement a full-fledged information aggregation tool set. That would mean not only combining data migration/mass download, replication/ETL, data warehouse/data mart, enterprise and mid-tier operational data store, and EII tools, but also defining — and aiding customers in defining — when to use each of these tools for strategic information management.

In most cases, the pioneers of new computing technologies take the arrows. Because the component technologies are already in most cases field proven and well supported, all that the pioneers of strategic information management risk is a relatively small added investment. By contrast, the benefits to the first mover are not only immediate competitive advantage but also short- and long-term cost savings plus long-term entrenched competitive advantage.

Chapter Five:

Market Forecast and Segmentation

Key Technologies for EII Initiatives

As stated in Chapter Two, EII is a business issue and, if implemented correctly, should improve the overall bottom line. Moreover, taking the appropriate first steps on an EII project can have a significant, positive effect on the final success of an entire information integration initiative.

By implementing EII solutions, enterprises can isolate their applications from the extensive differences in back-end databases and understand the potential of an open, heterogeneous, multisupplier computing infrastructure.

Enterprises are increasingly adopting EII-type approaches as part of their efforts to identify and rationalize their information assets. Earlier efforts to create a “universal repository” containing metadata about all of a corporation’s data, typically as part of a data warehousing effort, failed because lines of business had no incentive to hand over control of their information assets to a corporate IT department. In contrast, today’s lines of business clearly have a strong bottom-line incentive to make their information available to new customers via the Web, partners and suppliers via EBI, and internal employees via employee self-service portals.

The problem to date has been that information integration approaches have been fragmented, sometimes incompatible, and typically not so cost-effective as an overall EII solution.

Aberdeen EII Technology Segmentation Model

The Aberdeen EII segmentation model represents the various technology components that often complement an EII tool in an overall solution. The aim of the model is to identify suppliers that are most likely to complement an EII solution. Aberdeen defines the various EII “complementary technologies” in order to provide a clear understanding of the functionality provided by each technology category.

Aberdeen research indicates that integrating an EII solution with related technologies will be the most attractive integration solution for enterprises that seek to integrate their information in a global, cost-effective way.

Aberdeen notes the following complementary technologies:

- EAI technologies
- Database technologies
- Application server technologies
- XML messaging technologies

EAI

EAI suppliers' products offer the ability to pass data between enterprise applications. Their key technologies complementary to EII are data transport, data integration, and business process support.

Data Transport

In an EAI solution, data transport technology ("adapters" or ETL) provides support for communications between objects and applications, providing greater performance than standard file transfers. In addition, it provides flexibility in routing data or objects to many recipients on multiple platforms, regardless of system or network failures. Thus, EAI data transport can be useful in transporting data between an EII solution and an enterprise application or in adding application-specific data to the EII solution's information base.

Data Integration

In an EAI solution, data integration technology provides the ability to manage a disparate set of application data sources and targets using a tool set. This tool set allows the user to seamlessly flow data from step to step in a process without having to go outside of the tool. A data integration product has the ability to manage an increased volume or number of similar objects — for example, the volume of data transformations or the number of users accessing a data mart. The scalability of the data integration tool is linear — if the volume of data transformations doubles, then the time to process the data integrations should also double. A data integration product can manage a nonlinear increase in difficulty — measured in the workload of individuals or systems or in the functional limitations of software — because of new intricacies when working with the interrelationships between or among objects.

The application of EAI data integration tools in EII is straightforward. Therefore, EAI data integration can serve as an alternative way to combine data sources.

Business Process Support

A business process flow is inherently composed of multiple, distinct, work-based processes or steps. Usually, it is carried out over or through multiple organizations and managerial levels within an en-

terprise. Conceivably — and usually — not all of the personnel and organizations involved in the process will be users of the same line of business. The best process flow will need to reach out beyond its normally associated application boundaries to provide for enterprisewide process-flow automation.

In an EAI solution, support and integration are done at the level of business processes, not transactions. The solutions enable the easier reuse of the integration techniques across heterogeneous environments; organizations can also implement integration solutions without needing detailed knowledge of the application infrastructure's specific underlying technologies — a likely situation when the other application resides in a different company. This capability can be particularly useful when meshing an EII solution with the business processes in another company via EBI.

Database Technologies

Two sets of database supplier technologies are particularly useful as a complement to EII:

- Open veneers
- Integrated file and multimedia storage capabilities

Open Veneers

Sybase offers a product that provides a common front end to its own and other relational databases. This “veneer” lets developers write to a common SQL and allows users to issue the same transaction to any relational database. In combination with the integrated file and multimedia storage capabilities of suppliers such as Oracle, it allows relational access to a wide variety of data types. It does not allow the combination of data or a transaction to cross multiple suppliers' databases.

An open veneer such as Sybase's can significantly ease the EII solution's task of accessing multiple databases.

Integrated File and Multimedia Storage Capabilities

IBM/Informix (IDS), Oracle (Oracle8i and Oracle9i), and Microsoft (SQL Server 7.0 and 2000) have made a particular point of allowing users to fold both multimedia/object data and desktop files into their relational databases. IBM DB2 and Sybase Enterprise Server also provide some nonrelational-data-type and file access capabilities.

These capabilities allow the user to fold nonrelational data into an existing database in order to give the EII solution easier access to a wide variety of information and to provide greater ability to combine

that information by using the database's cross-data-type-transaction capabilities.

XML Messaging Technologies

XML is a descendant of SGML (Standard Generalized Markup Language) and a sibling of HTML (HyperText Markup Language) — both ISO (International Standards Organization) and Internet standards. XML was developed to add data management features similar to those of SGML, but went beyond HTML by keeping the data and its context or metadata together in the same “message.” As a result, when the information displayed on a Web page is stored in XML, a price change can be inserted on a Web page immediately by changing the data portion of the XML, without reprogramming the entire page.

XML also provides great improvement in the portability of data, inside and outside of the enterprise. By agreeing to accept the definition of data embodied in an XML message, another department, an enterprise portal, or an outside supplier can access, display, or respond to that data. To ease the use of this type of EBI in particular industries, an abundance of XML dialects (properly termed “vocabularies”) have emerged.

XML databases store data as XML objects in an object-oriented database. Although XML databases do not replace enterprise-scale databases, they are useful not only for permanent or “cache” storage of XML data but also as a repository for XML information used in XML-based interactions; as a second-tier or “cache” server for general-purpose Web data processing; and as a “common language” database for storing a wide variety of data types, such as files, multimedia data, and relational data. In such small-to-medium scale database situations, a typical advantage includes the storing of data in a widely useful format, supporting key XML-using applications with less customization and providing greater processing speed for transactions involving XML.

XML messaging is, therefore, a highly flexible way for the EII solution to communicate with both front-end programs/users and back-end databases. Moreover, an XML database can be a high-performance way for the EII solution to store cache data and directory metadata.

Combinations of Technologies

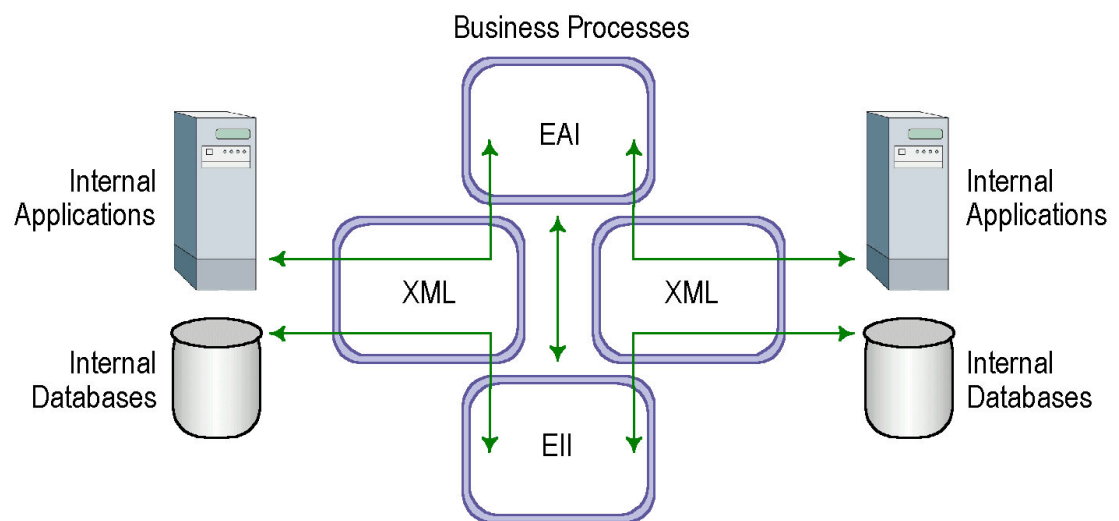
Combinations of these technologies are often useful in particular EII-related situations. Consider, for example, the products of Progress Corp. The ObjectStore XML database can be used as a mid-tier ODS. The PeerDirect InnerEdge and OuterEdge Server replication solution can deliver high performance in ETL and data migra-

tion/download. The SonicMQ XML messaging products support enterprise and cross-enterprise XML messaging infrastructure, allowing the exchange of information in such applications as EBI. The Progress OpenEdge “embedded infrastructure” allows organizations to apply the foregoing products plus an application server and enterprise database in creating scalable industry-specific applications rapidly.

Summarizing the Model

Figure 3 shows Aberdeen’s EII technology segmentation model, which can be used to categorize various EII and related offerings based on technology and functionality. The model also illustrates the interdependence of the various technologies.

Figure 3: EII Technology Segmentation Model



Source: Aberdeen Group, July 2003

Table 6 describes how the suppliers profiled in this report fit within the Aberdeen segmentation model.

Table 6: EII and Related Suppliers

Aberdeen EII Segmentation Model	
Technology Segmentation	Examples of Suppliers
EAI technology	Actional, BEA, Celcorp, Crossweave, iWay, SeeBeyond, TIBCO, Viewlocity, Vignette, Vitria, webMethods, CNT, MITEM, Tempest, Candle, CommerceQuest, Compaq, Compuware, Darc Software, DataMirror, Dharma, ETI, IBM, Level 8 Systems, Neon Systems, OST Business Rules, Percussion, Scribe Software, Sybase, Taviz Technology, Transoft, EnterWorks, Mercator, PTC, Software AG, WRQ
Database technology	IBM, IBM/Informix, Microsoft, Oracle, Sybase
XML messaging	Progress/Excelon, Ipedo, Metapa, Software AG (database); ArborText, Altova, and SoftQuad (publishing); DataChannel, Sequoia Software, Sybase, Bowstreet, SilverStream, Oracle, and iPlanet (enterprise portal); webMethods, InfoTera, IBM, Sun, i4i, Whitehill, and Inera (EBI); Vitria, Contivo (EDI); HP, IBM, Sun, Microsoft (Web services)
EII solution	Attunity, Business Objects, Ipedo, XAware, MetaMatrix, Venetica, BEA, Nimble Technologies, IBM (Information Integrator)

Source: Aberdeen Group, April 2003

Table 7 categorizes the EII suppliers based on the core technology solutions that they provide.

Table 7: EII and Related Suppliers

Aberdeen EII Segmentation Model	
Technology Segmentation	Examples of Suppliers
EAI Technology	Actional, BEA, Celcorp, Crossweave, iWay, SeeBeyond, TIBCO, Viewlocity, Vignette, Vitria, webMethods, CNT, MITEM, Tempest, Candle, CommerceQuest, Compaq, Compuware, Darc Software, DataMirror, Dharma, ETI, IBM, Level 8 Systems, Neon Systems, OST Business Rules, Percussion, Scribe Software, Sybase, Taviz Technology, Transoft, EnterWorks, Mercator, PTC, Software AG, WRQ
Database Technology	IBM, IBM/Informix, Microsoft, Oracle, Sybase
XML Messaging	Progress/Excelon, Ipedo, Metapa, Software AG (database); ArborText, Altova, and SoftQuad (publishing); DataChannel, Sequoia Software, Sybase, Bowstreet, SilverStream, Oracle, and iPlanet (enterprise portal); webMethods, InfoTe-ria, IBM, Sun, i4i, Whitehill, and Inera (EBI); Vitria, Contivo (EDI); HP, IBM, Sun, Microsoft (Web services)
EII Solution	Attunity, Business Objects, Ipedo, XAware, MetaMatrix, Venetica, BEA, Nimble Technologies, IBM (Information Integrator)

Source: Aberdeen Group, April 2003

Growth Forecasts

Most of the suppliers profiled in this report do not break out revenues associated with “data integration” — a frequently used supplier term for what we now call EII. Moreover, for suppliers, the dividing line between EII, EAI, content management, and ETL tools for data warehousing is frequently not a clear and precise one. Aberdeen has assigned to EII only revenues in which the primary purpose of the EII solution is to integrate data across databases for delivery to targets other than data warehouses. This category includes a small proportion of the EAI and ETL markets, as well as projects that involve an enterprise portal or EBI. Revenues from suppliers that do not provide a full-fledged EII solution are not included.

Based on available data regarding supplier revenue, Table 5 represents current market revenue and projected market growth. Aberdeen believes that the overall market will grow by 80% in 2003, with growth being especially fueled by the need to rationalize existing e-business efforts. Web services will contribute significantly to EII revenues over the next year, and they will be increasingly important from 2004 onward.

These estimates are strongly dependent on the overall growth of the high-tech sector of the economy. Specifically, if enterprises continue to delay spending on new initiatives, such as business process automation, into the first quarter of 2004, these numbers will be overestimates. If enterprises advance new initiatives in the third quarter of 2003, then these numbers will be underestimates.

Table 8 shows that, as yet, no single supplier dominates the market. Typically, suppliers do not release revenue figures. Therefore, these Aberdeen estimates are based on the number of employees and the number of years that a particular product has been offered.

Table 8: EII Supplier Revenues (Aberdeen Estimates, Projected over 2003)

Supplier	Revenues
Business Objects	\$15 million
Attunity	\$4 million to \$7 million
BEA	\$25 million
XAware	\$4 million to \$7 million
IBM	\$60 million
MetaMatrix	\$10 million

Nimble Technologies	\$4 million to \$7 million
Venetica	\$7 million to \$10 million

Source: Aberdeen Group, July 2003

Aberdeen believes that the professional services needed to successfully implement an EII initiative will become more easily measurable as revenue over the next 6 to 12 months. However, Aberdeen's field experience indicates that, for every dollar spent on software application licenses, \$1.25 to \$1.50 is spent on professional services to implement integration of the application and the resultant data.

Table 9 shows our estimate of the size of closely related markets. As these markets expand and mature, their products increase the demand for EII as an integral part of the solution. Thus, EII's impact extends far beyond its own market.

Table 9: Revenues for Markets Related to EII

Markets	2002	2003	2004
EAI	\$2.2 billion	\$2.6 billion	\$3.0 billion
Content management	\$1.2 billion	\$1.6 billion	\$2.0 billion
Portal	\$1.5 billion	\$2.2 billion	\$2.8 billion
Total	\$4.9 billion	\$6.4 billion	\$7.8 billion

Source: Aberdeen Group, April 2003

Chapter Six:

Aberdeen Observations

Impact of Major IT Trends

Aberdeen research shows that EII is on the critical path in corporate efforts to achieve comprehensive, information-leveraging, e-business architectures. Piecemeal efforts to link back-end databases to Web sites, EBI, and enterprise portals have left a rat's nest of point-to-point connections between Web server, application server, or applications and back-end Web text/graphics, workgroup files, or enterprise databases. EII's many-to-one-to-many approach is, therefore, well suited to rationalizing connections between internal and external users and key corporate information. However, its ability to do so is strongly affected by today's key IT trends, as described below.

Application Servers

Over the last two years, application servers have been steadily incorporating many of the middleware functions of an e-business architecture, including load balancing, acting as a development framework (hence, the increasing linkage of application servers and development tool sets), and application management. One of the main selling points of an application server has been that it is more manageable, scalable, and efficient because application servers can integrate data from a variety of heterogeneous sources, including back-office applications and databases. Therefore, over the next two years, many of today's EII solutions may possibly be incorporated in an application server — as BEA WebLogic for Liquid Data already does (Enosys). However, this might require the reprogramming of much of the existing code that is used to access back-end databases. Therefore, IT buyers examining an application server that incorporates an EII solution should carefully assess its ability to support existing database connections without recoding.

The Role of Java

Enterprise JavaBeans (EJB), the Java specification for server-side components, has received significant interest because of its ability to simplify and accelerate the development of business applications. By providing services such as transaction and resource management, persistence, mobility, versioning, scalability, and security, it enables the easier development of robust, mission-critical applications that access back-end databases — and using fewer developers. The wide availability of EJB 2.0 components means that programming data access in Java is becoming much easier, and EII is a complement that

makes the job easier still. In other words, because Java is well suited for EII and vice versa, the continued adoption of Java improves the market for and usefulness of EII.

Extensible Markup Language (XML)

XML is now a de facto, as well as du jour, standard. All major suppliers provide XML functionality, and many smaller suppliers focus on XML-based messaging, allowing enterprises to not only communicate externally but also drive business processes internally (and externally). As noted above, XML is particularly well suited for EII solutions because it can provide a common language for transmitting transactions from back-end data sources to internal and external users and programs. A proliferation of XML standards is causing difficulties for users in implementing XML connections with particular customers, partners, and suppliers, but that does not diminish the basic value proposition of EII. XML is, increasingly, a checklist item in EII solutions. A related upcoming standard is XQuery, which supports standardized querying on XML-formatted data.

Emphasis on Cost-Effective Administration

The economic downturn has caused enterprises to focus on rationalizing existing architectures for greater cost-effectiveness, rather than for building new e-business applications. A key area of concern is administration, as the costs of administrative personnel mount while hardware and networking costs decline. In response, suppliers have provided more automated and more administrator-friendly solutions. EII promises to reduce the administrative burden by offering a single administrative interface to multiple back-end databases, but, at present, the lack of administrative support in most EII solutions makes it appear that EII is an added cost burden rather than a cost savings. In fact, EII reduces administrative costs by simplifying user-to-database connections and making cross-database transactions more visible to the administrator. Over the next 18 months, increasing support for administration in EII products will allow administrators to begin to monitor multiple databases from different suppliers as one database. Moreover, connections to systems management tools will allow EII to take its proper place in overall systems administration.

Enterprise Portals

An enterprise portal allows the enterprise to coordinate the data displayed across applications and to fully leverage all major business-critical company databases. Moreover, a portal allows the enterprise to expose these data to new markets, such as Net markets and exchanges in general. Along with these competitive advantages come built-in cost savings — not only the cost savings derived from doing everything over the Internet (outsourced or consolidated) but also the

IT cost savings derived from simplifying the administration of the database/application connections, thereby saving key administrative costs. There is a side benefit of equal importance: e-information can unify the enterprise's "data islands." However, both of these "database benefits" will fail to emerge unless the enterprise portal uses EII technology to provide cross-database operations and displays. Therefore, EII is critical to the long-term success of enterprise portals. Likewise, a successful enterprise portal is a prime "target of opportunity" for users seeking to apply EII to initial projects promising major return on investment.

Web Services

Enterprises that take advantage of the Web in their businesses inevitably begin to implement Web services. The fundamental idea of Web services is to apply the tried-and-true idea of a service — a central piece of software that carries out specific functions that its users demand — to the Internet. The Web service is self-describing because it contains enough information about itself to allow any Web user or Web application to find out what it does and then to use it, making it a highly flexible service. The Web service's ability to use other Web applications to carry out complex tasks makes it feature rich and offers enterprises a new way to mine the Web for competitive-advantage applications and content.

Moreover, Web services allow IT to write one server version — no client version is needed — that can be automatically deployed across all the servers in the organization. It thereby becomes instantly available to the entire organization as well as global customers and a partner base of millions. These technical advantages also lead to broader business benefits, such as lowering the cost of doing business, allowing companies to achieve much greater efficiencies from using broadly available applications, and giving the organization greater ability to adapt quickly to changing conditions. These advantages are so great that Aberdeen expects implementing Web services and upgrading existing applications to be Web services will be the major new efforts of most enterprises over the next two years. EII plays a role in this effort because it provides a common access point from Web services to the "enterprise substrate" — key proprietary information. However, in order to allow organizations to fully leverage EII's potential in abetting Web services, EII solutions must provide SOAP support, frameworks supporting Web service development tools, and, if possible, a Web service provider interface. For further details, see the Aberdeen White Paper, *EII and Web Services* (July 2002).

Supplier Success Factors

To meet long-term user flexibility needs, suppliers should continue to integrate various middleware technologies into a smoothly working whole. The aim is to enable IT to create a distributed environment in which applications and data move instantly, cost-effectively, and flexibly to where a user needs them.

A supplier's EII solution should be extensible and reusable for the project at hand, as well as for future projects. The tools should be designed in such a manner that users are able to extend the tool from a per-project basis to ongoing enterprisewide usage.

In the long term, the EII supplier's tools and solutions should be able to contribute to an organization's bottom line by reducing the costs of application design, development, deployment, and, above all, maintenance. Suppliers should also satisfy overall user criteria, such as scalability and robustness.

Outside resources should be leveraged through strategic partnerships. A potential bottleneck for present EII suppliers could be customer demand for services. In-house services adapted to EII, key partnerships with systems integrators, as well as the professional services of major infrastructure suppliers, can alleviate this bottleneck. It is also important to build partnerships with suppliers of complementary software products (e.g., enterprise portal infrastructure suppliers).

Achieving customer satisfaction and delivering an expanding set of products will differentiate the new market leaders. Today, few EII providers have embraced integration with enterprise management products.

Suppliers should capitalize on increased interest in cost-effectiveness via solution consolidation. This trend provides EII suppliers with the opportunity to sell the ability to consolidate databases without the pain of physically merging them.

An unplumbed area is the use of EII for an "information audit." Service providers can use automatic "repository population" tools in EII solutions to walk into a customer site and gain a map of major data. This map, like asset management maps, can be highly useful in recommending ways of better leveraging information.

Success Factors for Organizations

Successful organizations establish business goals to be achieved by the implementation of an EII technology (e.g., increased sales and reduced cost of customer service). They identify the appropriate EII technology required to meet these goals and create metrics to gauge the successful use of the EII technology.

EII initiatives may involve integrating many sources and targets while maintaining independently coded modules for each integration process. This process is extremely resource intensive. Using a single, flexible, and consistent framework for each process will greatly reduce the amount of personnel resources needed to maintain and develop the system.

An EII solution should be built on an architecture that has the ability to scale, support, and introduce new technologies within a globally distributed environment. In addition, the solution should provide the flexibility to add multiple servers as a system grows, as well as the ability to use preconfigured components that “hook” into legacy databases.

Aberdeen advises IT buyers to demand that the EII suppliers define a path toward an integrated overall architecture. Choose the supplier with the best combination of a strong current solution, a good vision, and a solid, open position within the industry. Plan to prototype within the next six months and to spread EII across the enterprise architecture within the following year.

When selecting EII suppliers, organizations should look for two critical attributes:

1. Does the supplier have in-depth knowledge of the major databases to be integrated? Can the supplier demonstrate that the integration will be reliable, scalable, and possible? Is the supplier able to provide references that show systems in production?
2. Does the supplier have credibility (e.g., systems in production) in claiming to be able to support all of the relevant information integration initiatives in which your company expects to engage? Does the supplier have the consulting expertise or partnerships that can bring best practices from other integration implementations to bear on your business problems and opportunities?

The IT buyer should proactively involve users. The executives; business analysts; financial, marketing, and support representatives; and sales professionals who will eventually be using the information need to be involved in the integration requirements. These individuals can provide helpful insight into actual usage patterns and what will and will not be useful. Incorporating their opinions into the integration and deployment process also creates a sense of ownership and excitement on the part of users for the new integrated application systems.

Aberdeen Observations and Conclusions

Aberdeen research shows that there now exists a distinct market for EII that is beginning the “high-end user per-project” stage. This market is driven by enterprises’ needs to rationalize existing access to multiple databases, both to cut administrative costs and to leverage competitive-advantage, cross-database, proprietary information. EII solutions do this by providing many-to-one-to-many mechanisms to take a wide variety of back-end data sources, merge them into a common format, and pass them to a variety of front-end users and programs.

EII solutions also provide specific value-add to enterprise portal and EBI solution upgrade. Enterprise portals’ focus on providing end-users with access to multiple information sources means that EII’s ability to provide cross-database data and/or support cross-database transactions simplifies the developer’s job and the portal’s architecture. EBI solutions can use EII to expose more data to the outside, with more control over a single (EII) exposure point.

EII can also play an important role in Web services implementations. A “Web-servicized” EII solution can provide easier access by other Web services to local, remote, and cross-company data.

Positioning and Capabilities

Market Segmentation

As a new market with relatively few entrants, EII also has relatively few segments. The most meaningful division is between suppliers that focus primarily on EII (e.g., MetaMatrix) and suppliers for which “data integration” is one among many middleware products or that focus strictly on business intelligence (e.g., Business Objects). Some suppliers focus on EII for content databases; others focus on EII across data marts and decision-support databases.

As noted earlier, Aberdeen believes that the technology in the EII market is moving toward a full single database image or “veneer” that makes multiple databases look like a single database. As a result, suppliers that focus primarily on EII and can draw on knowledge of database technology are at a significant advantage in the next two years.

Technology Interdependency

Aberdeen research has indicated that one key criterion to being a long-time player in the EII market space is the ability to develop key alliances. These alliances provide complementary products, complementary services, efficient access to new markets, and joint efforts in development.

Aberdeen research indicates that EII suppliers will benefit most not from integration with each other's technologies, but via partnerships with full-function "infrastructure solutions" for particular enterprise needs. Again, the most attractive partnerships will allow better support for enterprise portals, EBI, and Web services. Other useful partnerships will combine EII and a mid-tier ODS, or EII and EAI.

This is clearly a window of opportunity for Web infrastructure suppliers such as Sun, IBM, BEA, and Sybase. By establishing partnerships with EII suppliers or acquiring EII solutions, these suppliers can position themselves as a stronger long-term solution in enterprise portals, EBI, and Web services. So far, IBM and BEA have been quickest to respond. By contrast, suppliers such as Oracle and Microsoft have not shown a full appreciation of the potential of EII.

EII suppliers have begun to establish alliances with professional service organizations (PSOs). The A-list comprises the Big Five PSOs (Andersen Consulting, Deloitte & Touche, Ernst & Young, KPMG, and PricewaterhouseCoopers) and industry service heavyweights (e.g., HP and Sun). Aberdeen expects the extension and deepening of those relationships over the next two years.

Distribution and Implementation

Many EII suppliers are young companies, in the range of one to four years. The market's awareness of most of these suppliers' EII identities began in 2001. Thus, the EII suppliers discussed here, except for Business Objects, BEA, and IBM, are still mostly in their formative stages.

Among these newcomers, there is no common distribution model for the EII market. All of these suppliers seek to define themselves as software solution suppliers that rely on their own systems integration efforts, third-party implementers, and training to address customer implementation requirements. Sales models are primarily direct, yet some companies have initiated alternative distribution or original equipment manufacturer (OEM) strategies as significant future distribution and revenue-enhancing strategies. Direct sales capabilities were limited at the start of 2003, but continue to significantly expand.

The established suppliers — IBM, BEA, and Business Objects — have already established channels, which they are beginning to leverage. Especially promising is the "services-led" model, in which EII comes bundled with extensive services for system integration and advice on strategic uses of EII and other software infrastructure tools and services.

Major Forces

Mergers and Acquisitions (M&A)

Because it is such a new and small market, most “newcomer” EII suppliers have not had the capital to pursue acquisitions. Aberdeen expects that over the next two years, most M&A activity will continue to center on larger “infrastructure solution” providers acquiring “newcomer” EII suppliers.

In the case of full-function EII solutions, a barrier to these acquisitions is that being acquired by one provider locks the EII solution supplier out of the installed base of other providers — for example, being acquired by IBM might lock an EII supplier out of Oracle’s installed base. However, because these installed bases are far larger targets of opportunity than EII suppliers currently have, Aberdeen expects some acquisitions to proceed despite the barrier.

Standardization and Integration

EII solutions are and must be firmly founded on standards such as SQL, ODBC/JDBC, and EJB. A key future standard is XML, and Aberdeen expects all EII suppliers to wrap their transactions in XML over the next two years, if they have not done so already.

As noted above, key targets for integration over the next two years will be application servers and Web services infrastructures. Application servers are already sources of legacy, cross-database code. In order to simplify the Web architecture, they will need to merge that code with code written directly on an EII framework. Some EII solutions that have not done so already will also need to define connections to Web services infrastructure solutions as another data “target,” as well as to define their solutions as Web services.

Success Factors

Suppliers must work toward achieving customer satisfaction. Delivering an expanding set of products will differentiate the new market leaders. Furthermore, when choosing an EII supplier, users should look for the critical attributes listed in *User Success Factors*. There were significant deployments in 2003, so IT buyers should expect to have enough reference accounts to interview. Organizations should apply the evaluation criteria cited in Chapter Two to ensure that they are implementing an integration platform that is flexible, extensible, and scalable.

The Future

Six enabling technologies will have a strong impact on EII solutions during 2003: application servers, Java, XML, database administra-

tion technologies, enterprise portals, and Web services. These technologies are critical to delivering long-term value-add from an EII implementation and to simplifying an e-business architecture using EII.

Organizations need to coordinate their initiatives for EII and EBI as well as EII and architecture rationalization or cost cutting. Doing so correctly will avoid another decade of slow, costly, and mostly unnecessary piece-by-piece data integration, and it will allow organizations to invest the capital saved into key new initiatives, such as Web services.

Chapter Seven: **Supplier Profile**

As noted above, many EII suppliers profiled in this report are small, recently created companies. The reader should, therefore, pay additional attention to their partnerships to determine their ability to support complex enterprise information architectures.

MetaMatrix, Inc.

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Market Position

MetaMatrix sees itself as a provider of an “open, rapid-deployment system” for helping enterprises “integrate data sources and manage metadata” globally. MetaMatrix aims its EII solution especially at:

- Financial services
- Government
- Migration services
- Large and midsize systems integration companies
- CRM/call center applications
- Enterprise portals
- ASPs
- Business intelligence, knowledge management, and data warehousing
- B-to-B and B-to-C exchanges
- Other custom applications

MetaMatrix’s full set of features, ability to tie in to formalized development methodologies, and administrative support make it a prime candidate for large enterprises with long-term EII, architecture rationalization, and administrative cost-cutting plans, or for enterprises seeking to rapidly develop or upgrade complex, business-critical, information-driven applications. MetaMatrix’s EII solution is also appropriate for a range of more focused efforts, including meshing with EAI solutions to coordinate business processes and business rules or providing dynamic querying across relational and rich-media “content” data such as Internet feeds or time series data, as well as supporting EBI, enterprise portals, and BI/Web analytics. Although MetaMatrix’s sophistication may make it “overkill” for

small organizations, its breadth of functionality makes it useful for most if not all EII efforts of large enterprises.

The MetaMatrix product set is a full-featured EII solution with MetaBase, a modeling tool based on MOF (OMG's Meta-Object Facility), with a searching and browsing tool (MetaViewer), a repository manager (MetaData Repository Manager), and MetaMatrix Server, with transaction support, development features, administrative tools, a "single database image," ETL "connectors," and access control, as well as publish/subscribe capabilities. MetaMatrix differentiates its EII solution especially by its:

- Abstraction Layer — MetaMatrix allows creation of virtual databases, which are a logical layer between the disparate data sources and the application developer or application. The layer integrates the information so that from the outside it looks like a single database to the developer or the application. Inside this layer, MetaMatrix metadata mappings point to the sources of each element.
- Two way XA-compliant transactions.
- Scalability owing to the solution to being entirely built in Java.
- Interoperability with a wide variety of front-end programs via XMI, JDBC, ODBC, SOAP, and XML support.
- Metadata management system based on standards such as MOF, XML Metadata Interchange (XMI), Common Warehouse Meta-Model (CWM), UML (Uniform Modeling Language), and other relevant OMG standards.
- A modeling tool that allows abstraction of existing metadata (improving programmer productivity by reducing custom coding), as well as creation of user-defined "metamodels."
- Sophisticated metadata-search, querying, and "connector" features.
- Ability to support business rules in the repository.

Core Products

MetaMatrix provides a "model-driven" EII solution that combines an enterprise metadata management system, MetaBase, with scalable query management technology. MetaMatrix offers two main products, MetaBase — which can stand alone as an enterprise metadata management system — and MetaMatrix Information Integration Server. MetaBase includes:

- MetaData Modeler, which uploads metadata from data sources, places them in a “model” format, and allows users to manage these models as well as creating models of their own
- The ability to create an abstraction layer through virtual metadata models
- MetaBase Repository, which houses physical and virtual models in an XMI format and uses them to generate runtime models for the MetaMatrix Information Integration Server
- Searchbase, a search tool that allows MetaData Modeler users to examine the contents of the MetaBase Repository in a flexible way
- MetaViewer, a browser-based search tool accessing metadata models and data types across intranets or the Internet

MetaMatrix Server includes:

- MetaMatrix Server, which provides user-definable virtual databases based on cross-database metadata, with views and support for SQL operations. MetaMatrix Server also allows developers to access data at a high “virtual metadata” level.
- MetaMatrix Administration Console, a GUI-based administrative console that monitors MetaMatrix Server operations, allows administration of user accounts and access rights, and provides a log of transactions.
- Query Builder, a utility that allows developers to create and test queries.
- MetaMatrix Connector Development Kit, an SDK that allows developers to create connectors and test them on a simulated MetaMatrix Server, as well as against live data.

MetaMatrix Server can be deployed on J2EE-compliant application servers such as IBM WebSphere, BEA WebLogic, SAP Application Server, and JBoss-compliant open source application servers.

MetaMatrix Server provides support for Web services via SOAP/XML, as well as access via ODBC and JDBC using SQL.

MetaMatrix’s EII solution allows cross-database access to relational and legacy databases, unstructured data, packaged-application databases, and XML-based messaging (e.g., data from across an organizational boundary). As noted above, it offers an extensible connector framework for development of custom connectors as well as support for JDBC-style connectors from a number of partners.

Services and Partnerships

MetaMatrix provides consulting and training in how to create and use its models. MetaMatrix partners with Consulting and Integration Partners to provide support throughout the customer's EII life cycle.

MetaMatrix recently announced an OEM agreement with SAP to provide MetaMatrix Server embedded in SAP NetWeaver, SAP's new Enterprise Services Architecture.

Aberdeen Conclusions

MetaMatrix's EII solution is the Cadillac of today's EII solutions — it is comprehensive in its feature set, able to support developers and administrators at a relatively high level, and includes sophisticated features for complex-application development. Thus, MetaMatrix may not be appropriate for medium-sized enterprises without the need for sophisticated modeling of multiple databases, but is highly suitable for the largest enterprises with complex information architectures or a need to develop complex cross-database applications.

Overall, MetaMatrix is clearly taking a leadership role among early entrants into the EII market in implementing the technology needed for EII to scale to the largest of enterprise needs and to provide a long-term enterprise information architecture. For these situations, IT buyers should look at MetaMatrix immediately.

Author Profile

Wayne Kernochan, Managing Vice-President, Platform Infrastructure

Wayne Kernochan focuses on databases, development environments, and proprietary hardware. His areas of coverage also include related software infrastructure, such as enterprise portal infrastructure, content management infrastructure, enterprise information integration solutions, and testing and software life cycle tools. He also contributes in related areas such as XML databases and tools, Web services, frameworks, and database administration utilities.

Kernochan has 25 years of experience in computers, including 12 as a programmer and technical manager in databases, development tools, e-mail, and office automation, most recently at Prime Computer. Before joining Aberdeen, Kernochan covered non-mainframe computer technologies for The Yankee Group.

Kernochan holds a B.A. in Government from Harvard University, an M.S. in Computer Science from Cornell University, and an M.B.A. in Information Systems from Sloan School of Management at M.I.T.

Appendix A:

Lexicon of Acronyms and Abbreviations

API	Application Program Interface
ASP	Application Service Provider
BI	Business Intelligence
CRM	Customer Relationship Management
EAI	Enterprise Application Integration
EBA	Enterprise Business Application
EBI	e-business Integration
EII	Enterprise Information Integration
EJB	Enterprise JavaBeans
ERP	Enterprise Resource Planning
ETL	Extract, Transform, Load
GUI	Graphical User Interface
HTML	Hypertext Markup Language
ITO	International Standards Organization
ITV	Independent Software Vendor
IT	Information Technology
J2EE	Java 2 Enterprise Edition
LAN	Local Area Network
LOB	Line-of-Business
OEM	Original Equipment Manufacturer
OLTP	Online Transaction Processing
PSO	Professional Service Organization
ROI	Return on Investment
SCM	Supply Chain Management
SDK	Software Development Kit
SGML	Standard Generalized Markup Language
SMB	Small to Medium-Sized Businesses

SNMP	Simple Network Management Protocol
SQL	Structured Query Language
TCO	Total Cost of Ownership
VAR	Value-Added Reseller
WAN	Wide Area Network
XML	Extensible Markup Language

Appendix B: **Related Aberdeen Research**

The following publication provides complementary market research relating to EII:

Embedded Infrastructure Cost of Ownership Study (December 2001)

Information on this and any other Aberdeen publication can be found at www.aberdeen.com or by e-mail at info@aberdeen.com.

Appendix C: **Supplier Contact Information**

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